Episode E0 | Overview

The nuances of the decisions of sellers behind the supply curve show us that market power breaks the efficiency of markets.

Introduction

In Part A we showed the core idea of economics, that specializing and trading could bend out the PPF, which is the technical way of saying that we can have more without needing more inputs of labor or resources and without needing better technology. Simply making better choices makes it possible to have more.

Then in Part B we showed that markets could facilitate the exchanges needed for specialization and trade. We also showed that this idea of social efficiency was maximized by markets, that anything more than what the market exchanged would be more costly than beneficial and anything less than what the market exchanged would be more beneficial than costly. So we actually want exactly what the market does.

But that was only in competitive markets, markets with many buyers and sellers, no externalities, and good information about the homogeneous good being exchanged.

In Part C we showed that externalities lead to misalignment between what the market delivers and what society would want. With negative externalities we want less than what the market delivers. With positive externalities we want more.

In Part D we just extended this idea of externalities a little further into markets with strategic interaction, where players' actions depend on what others are doing. We showed that the non-excludability of common resources and public goods in this light leads to the problems of over-extraction of common resources, much like with negative externalities, and the under-provision of public goods, much like with positive externalities.

Next we're turning to another way markets can succeed or fail, the market power of sellers. To do this, we need a more complete understanding of how sellers choose. How sellers choose starts with what they can make: how inputs like labor turn into output. I'll warn you, the next class is about the least interesting class of the semester for me. But that should tell you something, that if we're covering something as uninteresting as 'Costs', there's an even better reason why we're doing it.

Today we're going to run a simulation to collect data to build a more sophisticated model of the PPF, which will approach the bowed out nature of the PPF from another direction.

Simulation

We're going to make two goods in class today, Widgets (W) and Airplanes (A). We'll talk about the technical specifications for the two goods. To make these two goods I need 8 volunteers. You job will be turning these pieces of paper into the two goods.

- A Widget is made by stapling a piece of paper in half.
- An Airplane is a piece of paper that can be flown at least five feet.

As you're thinking about whether to volunteer, let me tell you a bit about the data we're about to collect. We're going to measure how many of each of these two goods we make with different levels of Labor (L) input into the production lines for Widgets (W) and Airplanes (A).

The table for this data will be a column for Labor for Widgets, then Quantity of Widgets, then the same for Airplanes.

L_W	Q_W	L_A	Q_A
8		0	
6		2	
4		4	
2		6	
0		8	

We'll start with a trial period where each gets a chance to learn how to make both W and A. Then start with all four making W and none making A for 20 seconds. Then two by two we'll switch to making A.

When everyone is done:

Plot both Production Functions. Talk about diminishing marginal returns. Talk about what happens when technology changes.

Plot the Labor curve, the relationship between inputs. Talk about being fundamentally constrained by the total labor.

Plot the PPF, the relationship between outputs.[1] Talk about the how the constraint on labor inputs gives us the frontier, and how the technology bends it out. And sometimes students will switch groups because they're good at airplanes, so that's specialization at work.

Next class we'll open with these numbers.

Materials: a stack of scrap paper, one stapler for the whole widget line, a five-foot line taped on the floor, and a timer. The single stapler is the fixed input. As more workers share it, each extra worker adds fewer widgets, which should make the flattening easier to see. It is also the capital that comes back in E1.

[1]This is where I've talked about the four quadrant PPF in the past. It gives a picture of the relationship between four different variables.